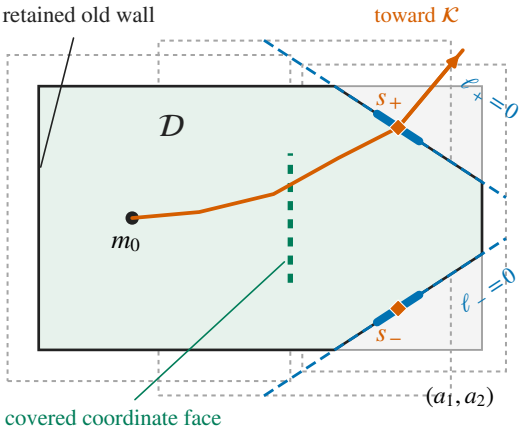
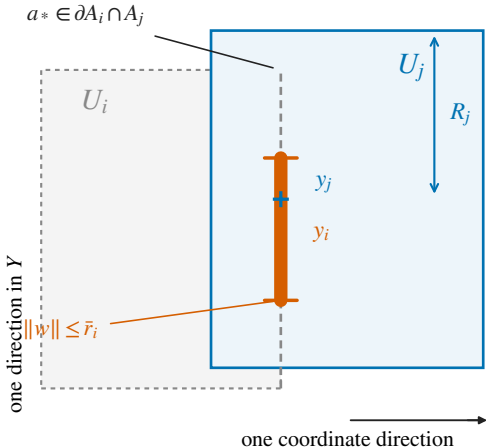
**(a) Exact cuts and coordinate coverage**




**(b) Entire low-sublevel fibers**




$$\bar{r}_i = \min(R_i, r_i^+), \qquad \|y_i - y_j\| + \bar{r}_i < R_j$$

Conceptual schematic: no calibrated coordinates or certified path data.